

✓ Task 3 : Data Visualization

Project Overview

This project focuses on presenting customer shopping data using different types of visualization . the main purpse is to make the information easier to understand and identify useful patterns from the dataset. In this task,I will use charts and graphs to visualize customer demographics , product categories, purchase amounts,ratings,payment methods and shopping frequency.

Objective

The objective of this task is to create clear and meaningful visualization that help in understanding customer shopping behaviour and comparing different aspects of the dataset.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

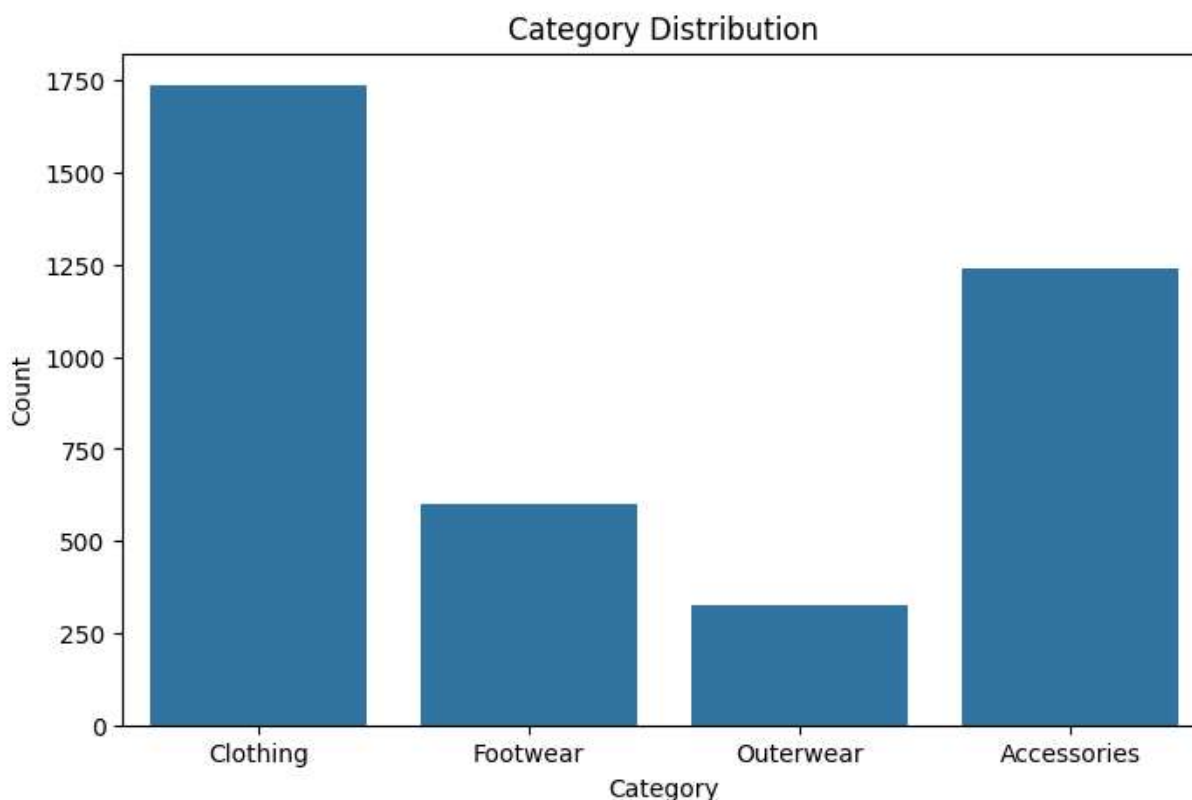
```
df = pd.read_csv("/content/shopping_trends.csv")
df.head()
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	
1	2	19	Male	Sweater	Clothing	64	Maine	L	M
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	M
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	M
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turc

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
plt.figure(figsize=(8,5))
sns.countplot(data=df , x="Category")
plt.title("Category Distribution")
plt.xlabel("Category")
plt.ylabel("Count")
plt.show()
```



✓ Observation

From the chart, I can see that Clothing has the highest number of purchases in the dataset. Accessories is the next most common category, while Footwear and Outerwear have fewer records.

This shows that the number of purchases is not the same across all product categories.

```
category_purchase = df.groupby("Category")["Purchase Amount (USD)"].mean()
print(category_purchase)
```

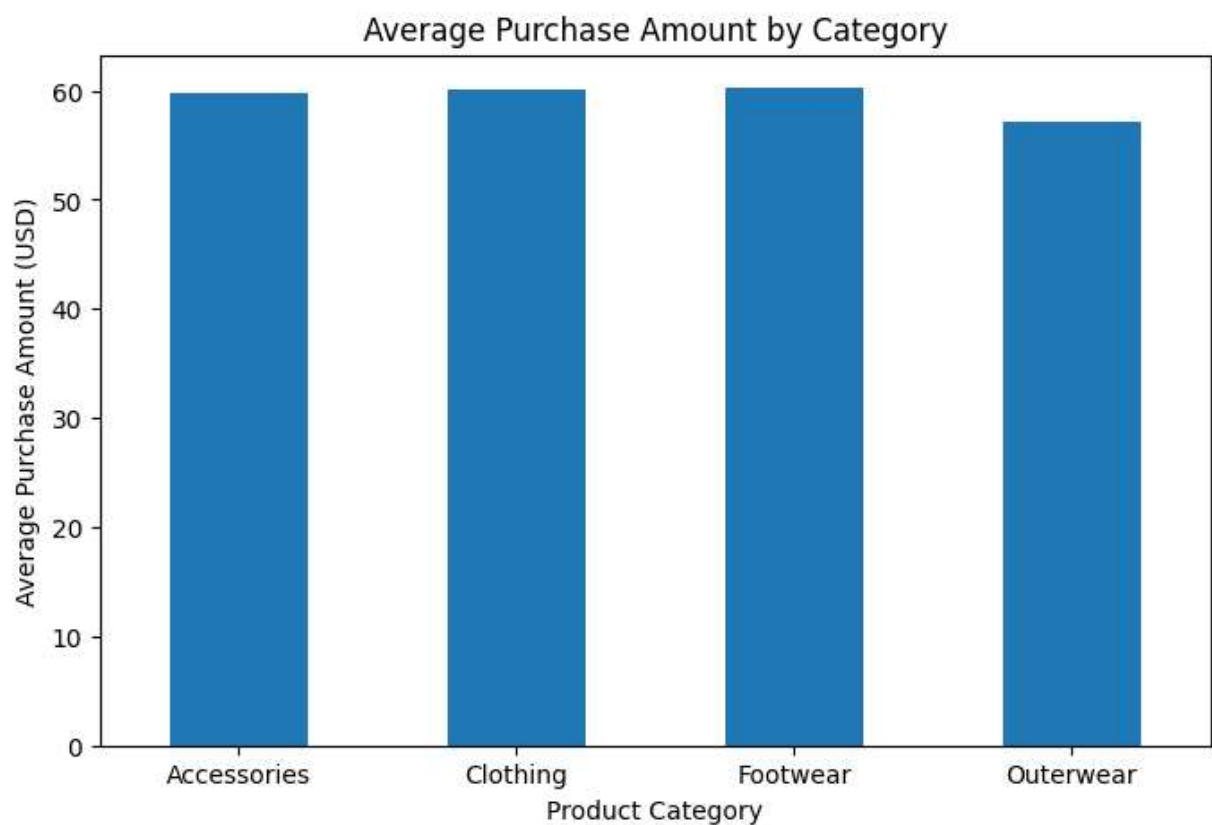
```
Category
Accessories    59.838710
Clothing       60.025331
Footwear       60.255426
Outerwear      57.172840
Name: Purchase Amount (USD), dtype: float64
```

```
plt.figure(figsize=(8,5))

category_purchase.plot(kind="bar")

plt.title("Average Purchase Amount by Category")
plt.xlabel("Product Category")
plt.ylabel("Average Purchase Amount (USD)")
plt.xticks(rotation=0)

plt.show()
```



✓ Observation

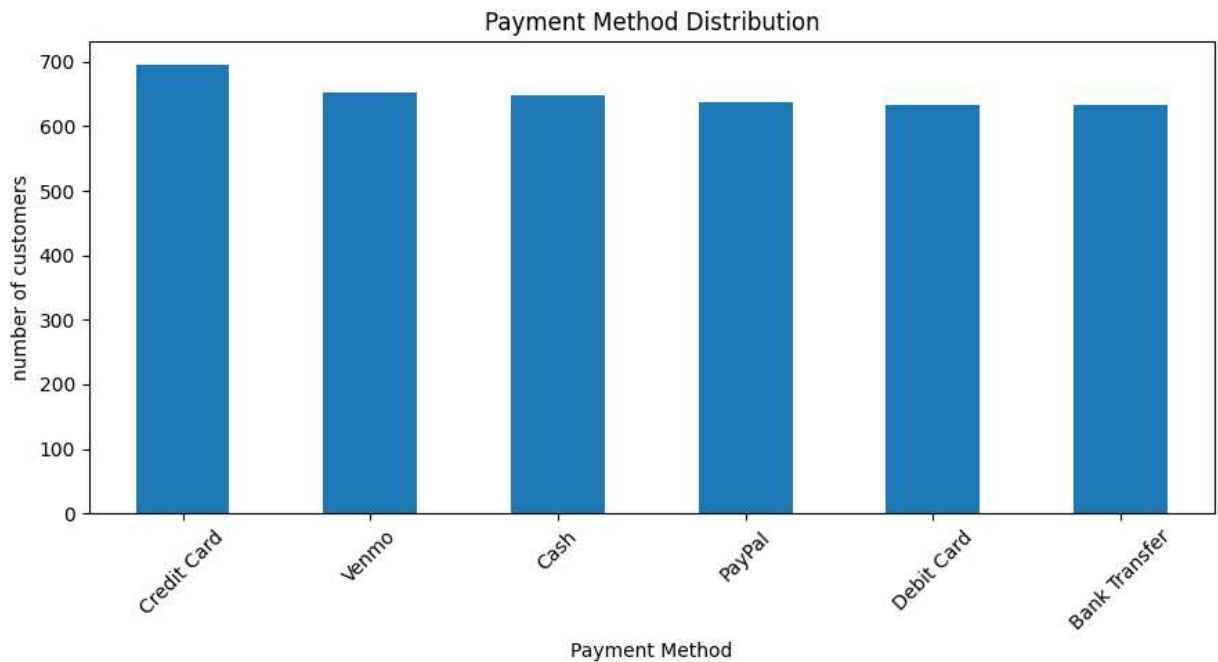
The average purchase amount is quite similar across all four categories. Footwear has the highest average purchase amount, while Outerwear has the lowest. The difference between the categories is not very large.

```
payment_counts = df["Payment Method"].value_counts()

print(payment_counts)
```

```
Payment Method
Credit Card      696
Venmo             653
Cash             648
PayPal           638
Debit Card       633
Bank Transfer    632
Name: count, dtype: int64
```

```
plt.figure(figsize=(9,5))
payment_counts.plot(kind="bar")
plt.title("Payment Method Distribution")
plt.xlabel("Payment Method")
plt.ylabel("number of customers")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



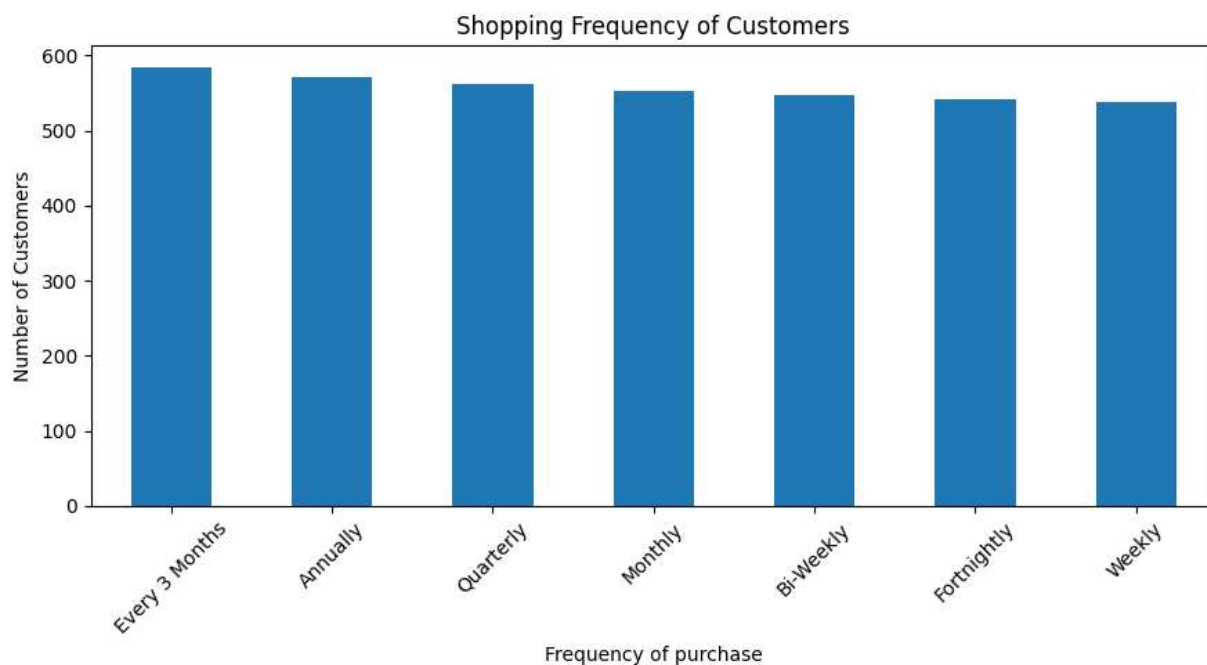
✓ OBSERVATION

From the chart, I can see that credit card is the most commonly used payment method in the dataset. Venmo and cash are also used by many customers, while the other payment methods have slightly fewer customers. Overall, the difference between the payment methods is not very large.

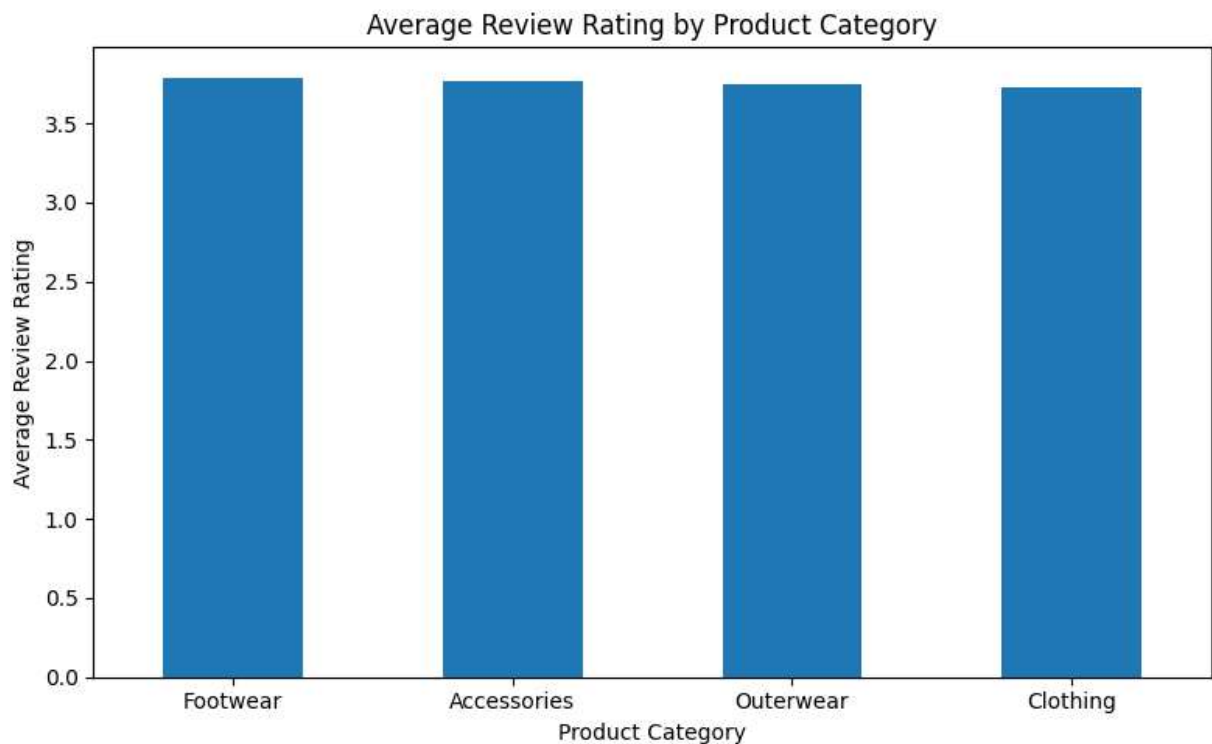
```
frequency_counts = df["Frequency of Purchases"].value_counts()
print(frequency_counts)
```

```
Frequency of Purchases
Every 3 Months    584
Annually          572
Quarterly         563
Monthly           553
Bi-Weekly         547
Fortnightly       542
Weekly            539
Name: count, dtype: int64
```

```
plt.figure(figsize=(9,5))
frequency_counts.plot(kind="bar")
plt.title("Shopping Frequency of Customers")
plt.xlabel("Frequency of purchase")
plt.ylabel("Number of Customers")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



```
category_rating = df.groupby("Category")["Review Rating"].mean().sort_values(ascending=True)
plt.figure(figsize=(8,5))
category_rating.plot(kind="bar")
plt.title("Average Review Rating by Product Category")
plt.xlabel("Product Category")
plt.ylabel("Average Review Rating")
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```



```
plt.figure(figsize=(8,5))

sns.scatterplot(data=df , x="Age" , y="Purchase Amount (USD)",alpha=0.5)
plt.title("Age vs Purchase Amount")
plt.xlabel("Age")
plt.ylabel("Purchase Amount (USD)")
plt.tight_layout()
plt.show()
```

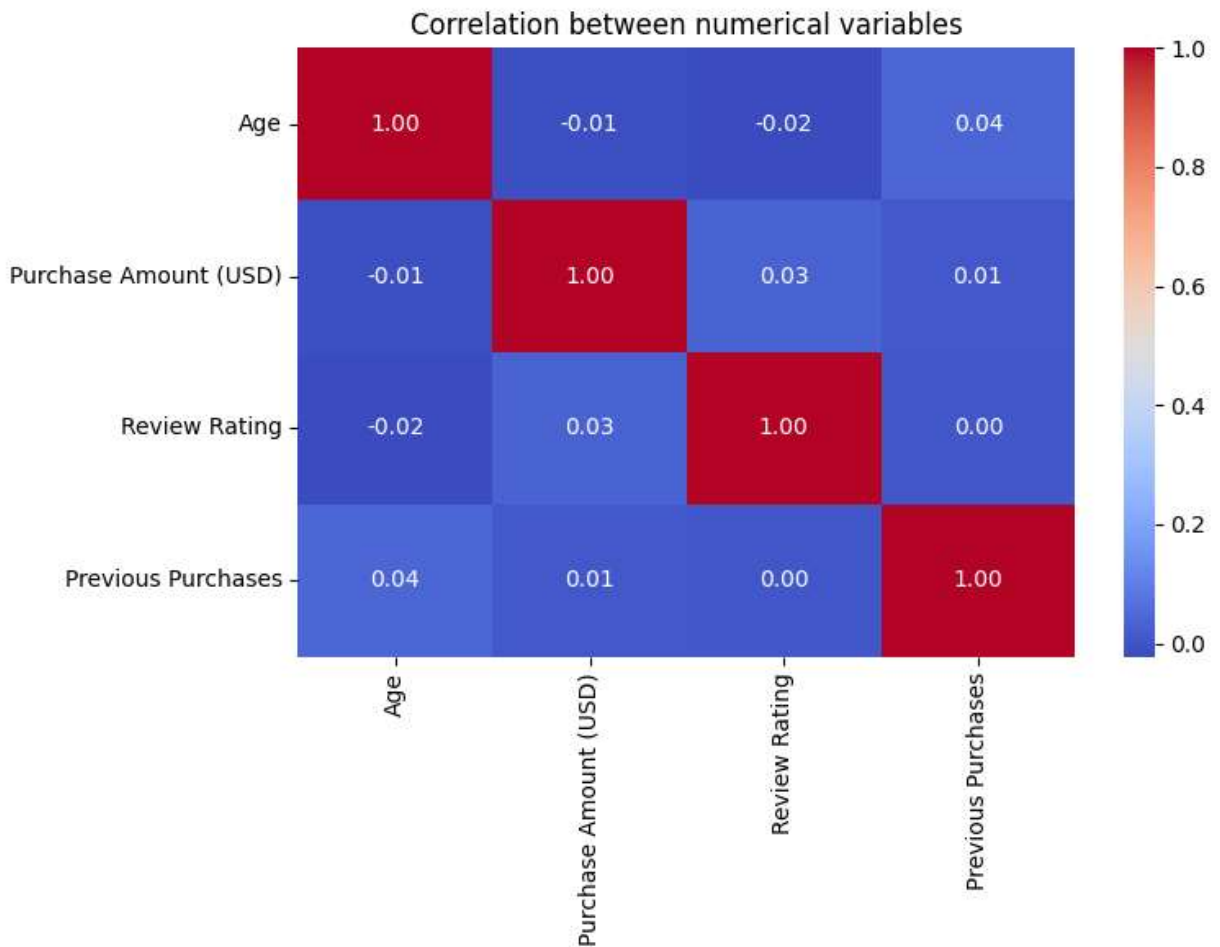


✓ OBSERVATION

The points are spread across different ages and purchase amounts without showing a clear pattern. This suggests that age does not have a strong visible relationship with purchase amount in this dataset.

```
numeric_columns =[
    "Age",
    "Purchase Amount (USD)",
    "Review Rating",
    "Previous Purchases"
]

Correlation_matrix = df[numeric_columns].corr()
plt.figure(figsize=(8,6))
sns.heatmap(Correlation_matrix,annot=True,cmap="coolwarm",fmt=".2f")
plt.title("Correlation between numerical variables")
plt.tight_layout()
plt.show()
```



Key Insights from Visualizations

The visualizations helped me understand the shopping behavior in the dataset more clearly.

- Clothing has the highest number of records among the product categories.
- The average purchase amount is quite similar across all product categories.
- Credit Card is the most frequently used payment method.
- Customer shopping frequencies are fairly balanced across the different categories.
- The Age vs Purchase Amount plot does not show a clear relationship between the two variables.